

Project Development Phase

Model Performance Test — Artificial Intelligence

Date: 29 June 2026

Team ID: N/A

Project Name: Network Anomaly Detection

Maximum Marks: 10 Marks

Model Performance Testing:

S.No.	Parameter	Values / Screenshot	Status
1.	Model Summary	NOT APPLICABLE This template is designed for deep-learning / computer-vision models that expose a <code>model.summary()</code> — typically Keras/TensorFlow sequential or functional architectures (e.g. CNNs, LSTMs, autoencoders). The Network Anomaly Detection project uses Isolation Forest, a classical unsupervised ensemble algorithm implemented in scikit-learn. It has no neural-network architecture, no layer graph, and no trainable-parameter count in the sense expected by this section. All classification performance metrics (accuracy, precision, recall, F1-score, confusion matrix) are documented in the Machine Learning Performance Test template.	N/A
2.	Accuracy Training Accuracy — Validation Accuracy —	NOT APPLICABLE Training accuracy and validation accuracy curves (epoch-by-epoch) are metrics specific to iteratively trained models (neural networks). Isolation Forest is fit in a single pass over the normal-traffic training set; it does not iterate over epochs and does not produce a training/validation split in the conventional sense. Post-hoc evaluation accuracy against the held-out labelled test set (79%)	N/A

S.No.	Parameter	Values / Screenshot	Status
		is reported in the Machine Learning template.	
3.	Confidence Score (YOLO Projects Only) Class Detected — Confidence Score —	NOT APPLICABLE Confidence scores and detected classes are metrics produced by object-detection models (e.g. YOLOv5, YOLOv8). The Network Anomaly Detection project performs binary traffic classification (Normal / Anomaly) on tabular network-flow features using Isolation Forest. It does not use YOLO or any object-detection framework.	N/A

Note: All quantitative model performance results are recorded in the Machine Learning Model Performance Test template, which is the appropriate template for this project.